



COMSATS University Islamabad

Department of Computer Science

Course Syllabus

Course Information

Course Code: **CSC301**

Credit Hours: **3(3,0)**

Lab Hours/Week: **0**

Course Title: **Design and Analysis of Algorithms**

Lecture Hours/Week: **3**

Pre-Requisites: **CSC211-Data Structures and Algorithms**

Catalogue Description:

This course is designed to provide knowledge of the principles and techniques used in the design and analysis of algorithms. Topics cover: Overview of Algorithm; Proving Correctness of Algorithms; Asymptotic Notations; Solving Recurrence Relations; Sorting & Order Statistics; Brute Force Algorithms & their Analysis; Divide and Conquer; Dynamic Programming; Greedy Algorithms; Graph; and Basic Computability.

Text and Reference Books

Textbook:

1. Introduction to the Design and Analysis of Algorithms, Levitin, A., Pearson, 2017.

Reference Book:

1. Introduction to Algorithms, Cormen, T. H. Leiserson, C.E., Rivest, R.L. & Stein, C., MIT Press, 2019.

Week wise Plan:

Lecture	CDF Unit #	Topics Covered	Reading Material
1.	1	The concept: What are Algorithms, Why Do You Need to Study Algorithms, What Kinds of Problems are Solved by Algorithms, Algorithms as a Technology, How to Represent an Algorithm; Properties of Algorithms: Input, Output, Precision, Finiteness, Definateness, Correctness, and Generality.	CLRS: Ch1 Levitin: Ch1
2.	1	Fundamentals of Algorithmic Problem Solving: The STAIR Steps for Solving Problems, and Major Factors in Designing an Algorithm.	Levitin: Ch1
3.	2	Pre-condition, Post-condition, Partial Correctness of an Algorithm, Total Correctness of Algorithm, and Illustrative Examples.	Rosen: Ch5
4.	2	Loop Invariants, Correctness of Iterative Algorithm: Initialization, Maintenance, and Termination.	CLRS: Ch3 Thomas: Ch4
5.	2	Correctness of Recursive Algorithm: Quick Review of Mathematical Induction, Proving Correctness of Recursive Algorithm using Induction, and Illustrative Examples.	Thomas: Ch4
6.	3	Time Complexity Measuring Notations: Growth Rate of Functions; Asymptotic Notations: Big O Notation, Sigma Notation, and Theta Notation.	CLRS: Ch3 Levitin: Ch2
7.	3	Major Assumptions in Analyzing Algorithms, Model of Computation (RAM Model), Mathematical Analysis of Non-Recursive Algorithms, and Worst, Best & Average Case Behavior of Algorithms.	Levitin: Ch2
8.	3	Complexity Classes i.e., Constant, Linear, Quadratic; Analysis of Iterative Algorithms, Empirical Measurements of Performance; and Time & Space Tradeoffs in Algorithms.	CLRS: Ch3 Levitin: Ch2
9.	4	Mathematical Analysis of Recursive Algorithms, and Solving	CLRS: Ch4

		Recurrences: Substitution Method.	Levitin: Ch2
10.	4	Recurrence Tree Method.	CLRS: Ch4 Levitin: Ch2
11.	4	Master Theorem for Solving Recurrences.	CLRS: Ch4 Levitin: Ch2
12.	5	Merge Sort: Overview, Worst, Best & Average Case Analysis; Quick Sort: Overview, and Worst, Best & Average Case analysis.	CLRS: Ch4,Ch7 Levitin: Ch4
13.	5	Randomized Quicksort; Heap Sort: Overview, and Worst, Best & Average Case Analysis.	CLRS: Ch7, Ch6 Levitin: Ch6
14.	5	Sorting in Linear Time: Lower Bounds for Sorting, Counting Sort, and Radix Sort.	CLRS: Ch8 Levitin: Ch7
15.	6	Brute Force Algorithms & their Analysis: Pattern Matching Algorithm & its Time Complexity.	Levitin: Ch3 CLRS: Ch3
16.	6	Brute Force Algorithms & their Analysis: Finding Inversion in an Array, and Closest-Pair & Convex-Hull Problems.	Levitin: Ch3
17.	Mid Term Exam		
18.			
19.	7	Dynamic Programming: Comparison of DP, Divide & Conquer, and Elements of Dynamic Programming.	CLRS: Ch15 Levitin, A: Ch8
20.	7	Dynamic Programming (Matrix Chain Multiplication): Problem Analysis, Notations, Designing DP Algorithm for MCM & its Time Complexity, and Applications of MCM.	CLRS: Ch15
21.	7	Dynamic Programming: Optimal Binary Search Trees & its Time Complexity.	CLRS: Ch15 Levitin: Ch8
22.	7	Dynamic Programming (0/1 Knapsack Problem): Problem Analysis, Notations, Designing DP Algorithm for 0/1 Knapsack Problem & its Time Complexity, and Applications of 0/1 Knapsack Problem.	CLRS: Ch15 Levitin: Ch8
23.	7	Dynamic Programming (Longest Common Subsequence): Problem Analysis, Notations, Designing DP Algorithm for LCS & its Time Complexity, and Applications of LCS.	CLRS: Ch15
24.	7	Dynamic Programming (Edit Distance Problem): Edit, Edit Distance Designing DP Algorithm for Edit Distance Problem & its Time Complexity, and Applications Analysis of DP Edit Distance.	CLRS: Ch15
25.	8	Greedy Algorithms: Huffman Encoding & its Time Complexity.	CLRS: 16 Levitin: Ch9
26.	8	Greedy Algorithms: Activity Scheduling, Disjoint Subsets & Union-Find Algorithms.	CLRS: 16
27.	9	Graphs: Terminology, Representation Techniques, Adjacency Matrix, Adjacency List; Various Applications of Graphs, Elementary, Topological Sorting using DFS, and Strongly Connected Component.	CLRS: Ch22
28.	9	Warshall's Algorithm, and Floyd's Algorithm for the All-Pairs Shortest-Paths Problem.	CLRS: Ch24 Levitin: Ch8
29.	9	Maximum Flow Problem & its Working.	CLRS: Ch24 Levitin: Ch10
30.	9	Time Complexity of Maximum Flow Algorithm.	CLRS: Ch24

			Levitin: Ch10
31.	10	Introduction to Complexity Classes & Computability.	CLRS: Ch34 Levitin: Ch11
32.	10	NP Complete Problem.	CLRS: Ch34 Levitin: Ch11
Final Term Exam			

Graduate Attributes (GAs)

S.#	Description
2	Apply knowledge of computing fundamentals, knowledge of a computing specialization, and mathematics, science, and domain knowledge appropriate for the computing specialization to the abstraction and conceptualization of computing models from defined problems and requirements
3	Identify, formulate, research literature, and solve <i>complex</i> computing problems reaching substantiated conclusions using fundamental principles of mathematics, computing sciences, and relevant domain disciplines
4	Design and evaluate solutions for <i>complex</i> computing problems, and design and evaluate systems, components, or processes that meet specified needs with appropriate consideration for public health and safety, cultural, societal, and environmental considerations

Course Learning Outcomes (CLO)

Sr.#	Unit #	Course Learning Outcomes	Blooms Taxonomy Learning Level	GA
CLO-1	1-2	Demonstrate an algorithmic approach to a given problem.	<i>Understanding</i>	2
CLO-2	3-7	Design new algorithms for different computational problems.	<i>Creating</i>	2-4
CLO-3	8	Prove correctness of an algorithm using loop invariant and induction.	<i>Applying</i>	2
CLO-4	9	Analyze best, average, and worst-case behaviors of an algorithm.	<i>Analyzing</i>	2
CLO-5	10	Explain the concept of various complexity classes with examples.	<i>Understanding</i>	2

CLO Assessment Mechanism

Assessment Tools	CLO-1	CLO-2	CLO-3	CLO-4
Quizzes	Quiz 1	Quiz 2	Quiz 3 &4	-
Assignments	Assignment 1	Assignment 2	Assignment 3 &4	-
Mid Term Exam	Mid Term Exam	Mid Term Exam	-	-
Final Term Exam	Final Term Exam			

Policy & Procedures

- **Attendance Policy:** Every student must attend 80% of the lectures as well as laboratory in this course. The students falling short of required percentage of attendance of lectures/laboratory work, is not allowed to appear in the terminal examination.

- **Course Assessment:**

	Quizzes	Assignments	Mid Term Exam	Terminal Exam	Final Marks
Theory (T)	15	10	25	50	100

- **Grading Policy:** The minimum passing marks for each course is 50% (In case of LAB; in addition to theory, student is also required to obtain 50% marks in the lab to pass the course). The correspondence between letter grades, credit points, and percentage marks at CUI is as follows:

Grade	A	A-	B+	B	B-	C+	C	C-	D+	D	F
Marks	>= 85	80 - 84	75 - 79	71 - 74	68 - 70	64 - 67	61 - 63	58 - 60	54 - 57	50-53	< 50
Cr. Point	3.67-4.00	3.34-3.66	3.01-3.33	2.67-3.00	2.34-2.66	2.01-2.33	1.67-2.00	1.31-1.66	1.01-1.30	0.10-1.00	0.00

- **Missing Exam:** No makeup exam will be given for final exam under any circumstance. When a student misses the mid-term exam for a legitimate reason (such as medical emergencies), his grade for this exam will be determined based on the Department policy. Further, the student must provide an official excuse within one week of the missed exam.
- **Academic Integrity:** All CUI policies regarding ethics apply to this course. The students are advised to discuss their grievances/problems with their counsellors or course instructor in a respectful manner.
- **Plagiarism Policy:** Plagiarism, copying and any other dishonest behaviour is prohibited by the rules and regulations of CUI. Violators will face serious consequences.